

Khushal Midha

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EDUCATION

Indian Institute of Information Technology, Lucknow
B.Tech in Computer Science and Artificial Intelligence – CGPA: 8.72/10

Courses: OOPS, DBMS, OS, DSA
August 2023 – July 2027

EXPERIENCE

SDE Intern – Coffeee.io Remote — Jun 2025 – Aug 2025

- Built a **project-based assessment module** (Java, Spring Boot, React, PostgreSQL) where candidates debug live codebases via **error logs**, graded automatically against hidden test cases.
- Developed a **read-only AI assistant** (LLMs, Redis, Docker) to provide contextual hints without edit access, preserving evaluation integrity; validated in **staging**.

PROJECTS

- 1. MediPulse** Feb 2025 – Present
MERN, FastAPI, Hugging Face Transformers, Redis, Kafka, Socket.IO, WebRTC, RAG Live Demo </>
 - Built a **multi-tenant healthcare SaaS** with **125 REST APIs**, **24 MongoDB models**, and **34 React pages**, supporting **appointments**, **live OPD queues**, **WebRTC consultations**, and **hospital portals** with **7-role RBAC** via **JWT & Google OAuth**.
 - Developed an **AI triage engine** combining a **Gemini agent**, **3 Hugging Face models**, and a custom **RAG pipeline** (using **text-embedding-004 & Cosine Similarity**) via **FastAPI** for accurate **disease prediction**, **specialty routing**, and **context-aware patient memory**.
 - Engineered **real-time OPD queues** and **virtual wallets** using **Socket.IO**, **Redis**, and **Kafka**, implementing **atomic transactions**, **distributed locks**, **idempotent refunds**, and **asynchronous event processing** for **concurrency-safe operations**.
- 2. ThreatGraph** Jan 2026 – Present
Python, FastAPI, PyTorch, Kafka, AWS EC2, Docker, React.js, PostgreSQL Live Demo </>
 - Engineered a **real-time threat intelligence platform** on **AWS EC2** via **Kafka** and **FastAPI**, streaming **10k+ network flows/sec** through a hybrid **Temporal GNN + Transformer** model detecting complex lateral movement attacks with **0.91 F1-score** and **sub-second latency**.
 - Architected **RAG-driven AI SOC investigations** mapped to **MITRE ATT&CK runbooks**, interactive **Force-Graph attack topologies**, and **automated zero-trust containment** via **host micro-isolation**.
- 3. TriageAgent** May 2026 - Present
Python, LiteLLM, FAISS, RAG, Sentence-Transformers, LLMs
 - Architected a **provider-agnostic AI support agent** utilizing **LiteLLM** and **FAISS** for semantic retrieval (RAG), improving intent classification accuracy by **37%** (achieving 81.4%) over keyword baselines.
 - Engineered a **6-signal risk escalation engine** and an automated **LLM-as-a-Judge** evaluation pipeline, achieving a perfect **1.0 Cohen's Kappa** agreement score and eliminating manual QA bottlenecks.

ACHIEVEMENTS

- Rated as **Expert Codeforces(1809)**, **5-star CodeChef (2137)**, **Guardian LeetCode (2137)**.
- Solved **3000+ DSA problems** across Codeforces and LeetCode strengthening problem-solving skills.
- Selected for **Amazon ML School 2025**, among top 3000 from 85,000 applicants Proof </>
- Secured rank of **186** in **Facebook Meta Hackercup Round-3** among 14000+ Participants Ranklist </>
- Qualified **Adobe Gensolve Hackathon**, ranked **top 5%** among 54000+ participants. Certificate </>
- ICPC Regionalist** at the **Amritapuri Regionals 2024-25** with team **Overshadowed**.
- Topped among **250+** peers in **Freshers Cup 2024** by CP Wing, Axios, IIIT Lucknow
- Achieved All India Rank **15,503** in **JEE Advanced 2023** and Rank **9,932** in **JEE Mains 2023**.

SKILLS & OTHERS

- Languages:** C, C++, Python, JavaScript, Java, SQL
- Frontend:** React.js, TypeScript, HTML, CSS, Tailwind CSS, Bootstrap
- Backend:** Node.js, Express.js, Spring Boot, Redis, Kafka, Fast API
- Database:** MongoDB, MySQL, PostgreSQL
- Machine Learning & AI:** PyTorch, XGBoost, Scikit-learn, Transformers, MLOps (MLflow), LLMs, Agentic AI, Prompt Engineering
- Courses:** Object Oriented Programming, Database Management System, Computer Networks, Operating System
- Technical Skills:** Data Structures Algorithms, Competitive Programming, Web Development, REST API Design